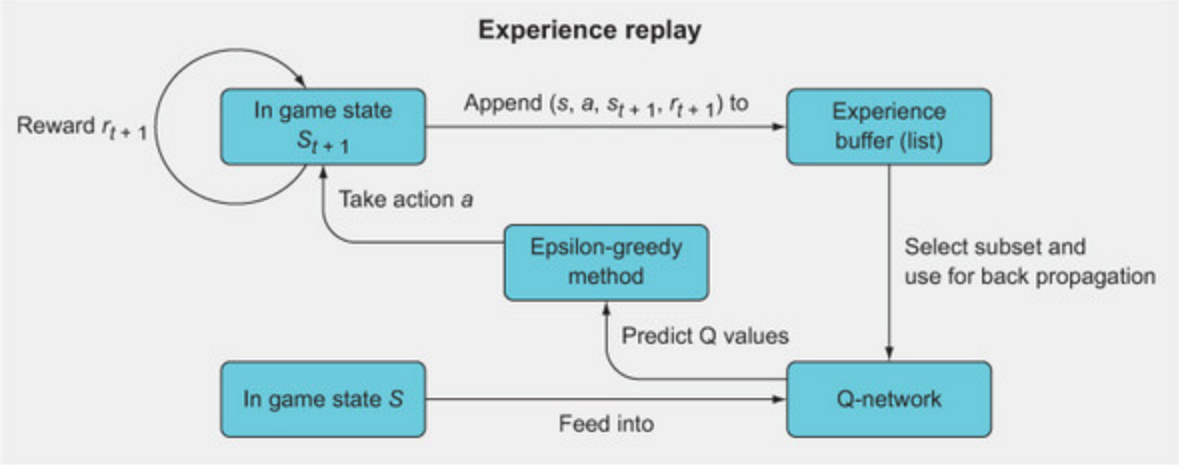


Thus, in addition to learning the action value for the action you just took, you're also going to use a random sample of past experiences to train on, to prevent catastrophic forgetting.

Listing 3.5 shows the same training algorithm from listing 3.4, except with experience replay added. Remember, this time we're training it on the harder variant of the game, where all the board pieces are randomly placed on the grid.

Listing 3.5. DQN with experience replay

```
from collections import deque
epochs = 5000
losses = []
mem_size = 1000
1
batch_size = 200
2
replay = deque(maxlen=mem_size)
3
max_moves = 50
4
h = 0
for i in range(epochs):
    game = Gridworld(size=4, mode='random')
    state1_ = game.board.render_np().reshape(1,64) +
np.random.rand(1,64)/100.0
    state1 = torch.from_numpy(state1_).float()
    status = 1
    mov = 0
    while(status == 1):
        mov += 1
```